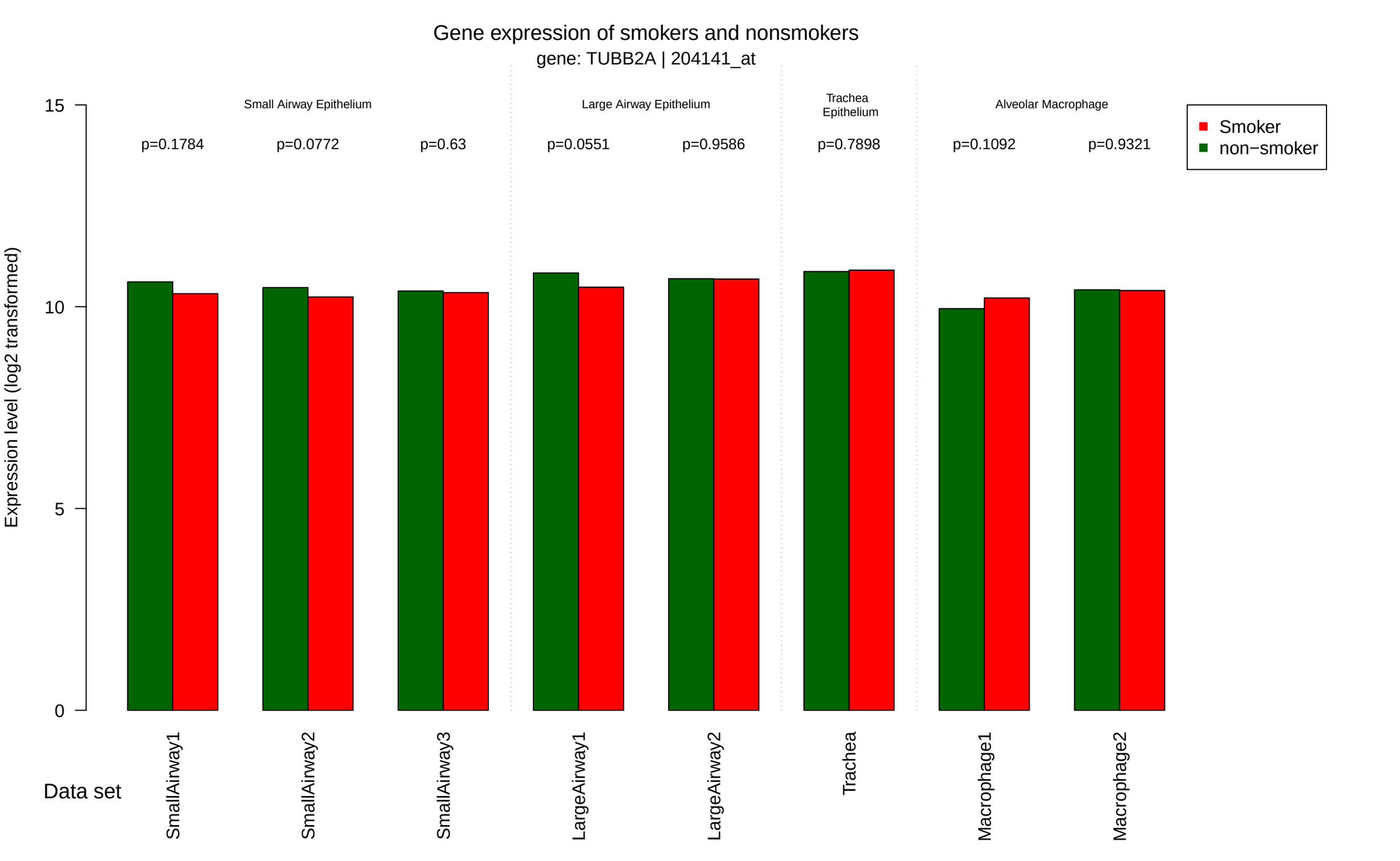
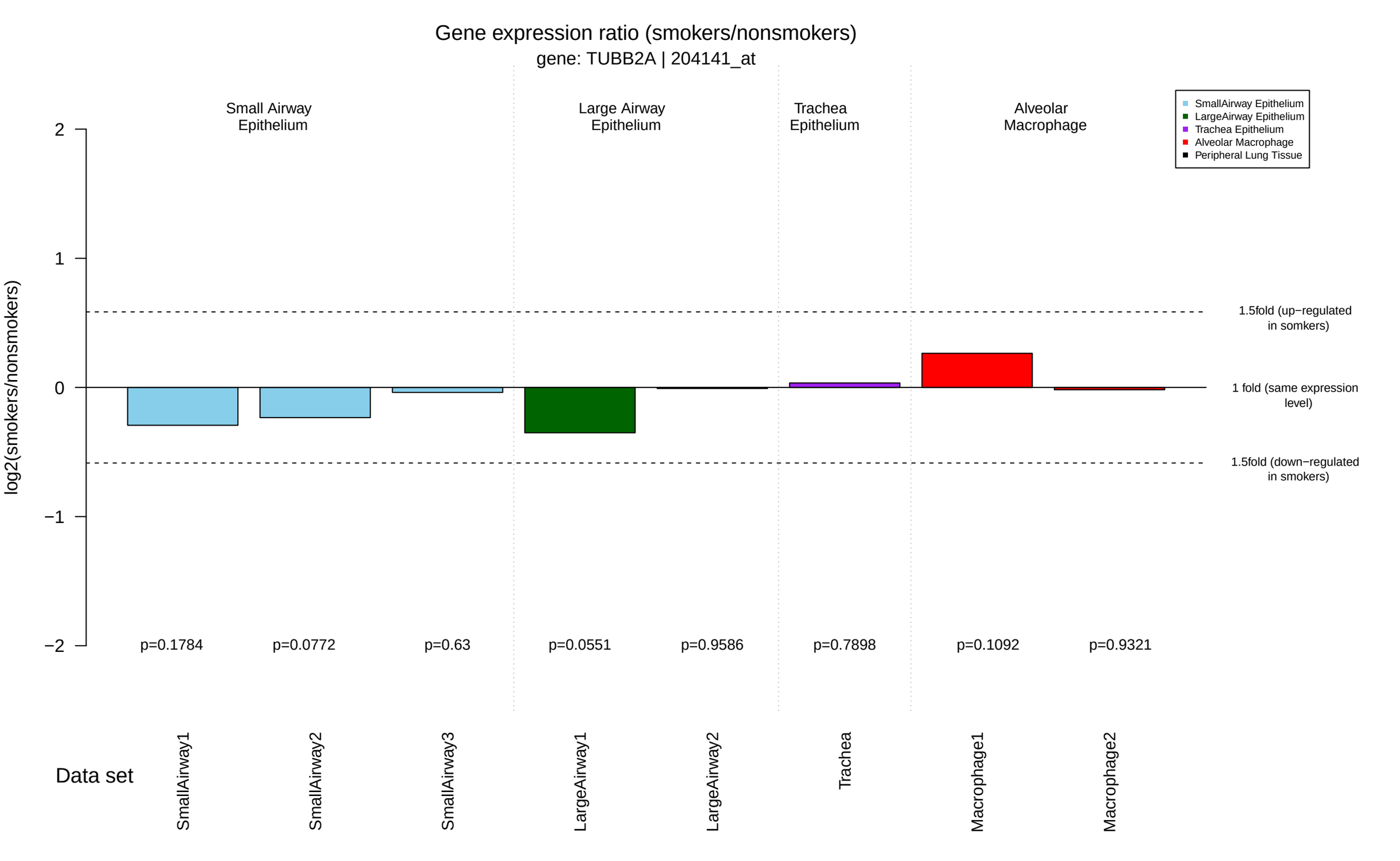
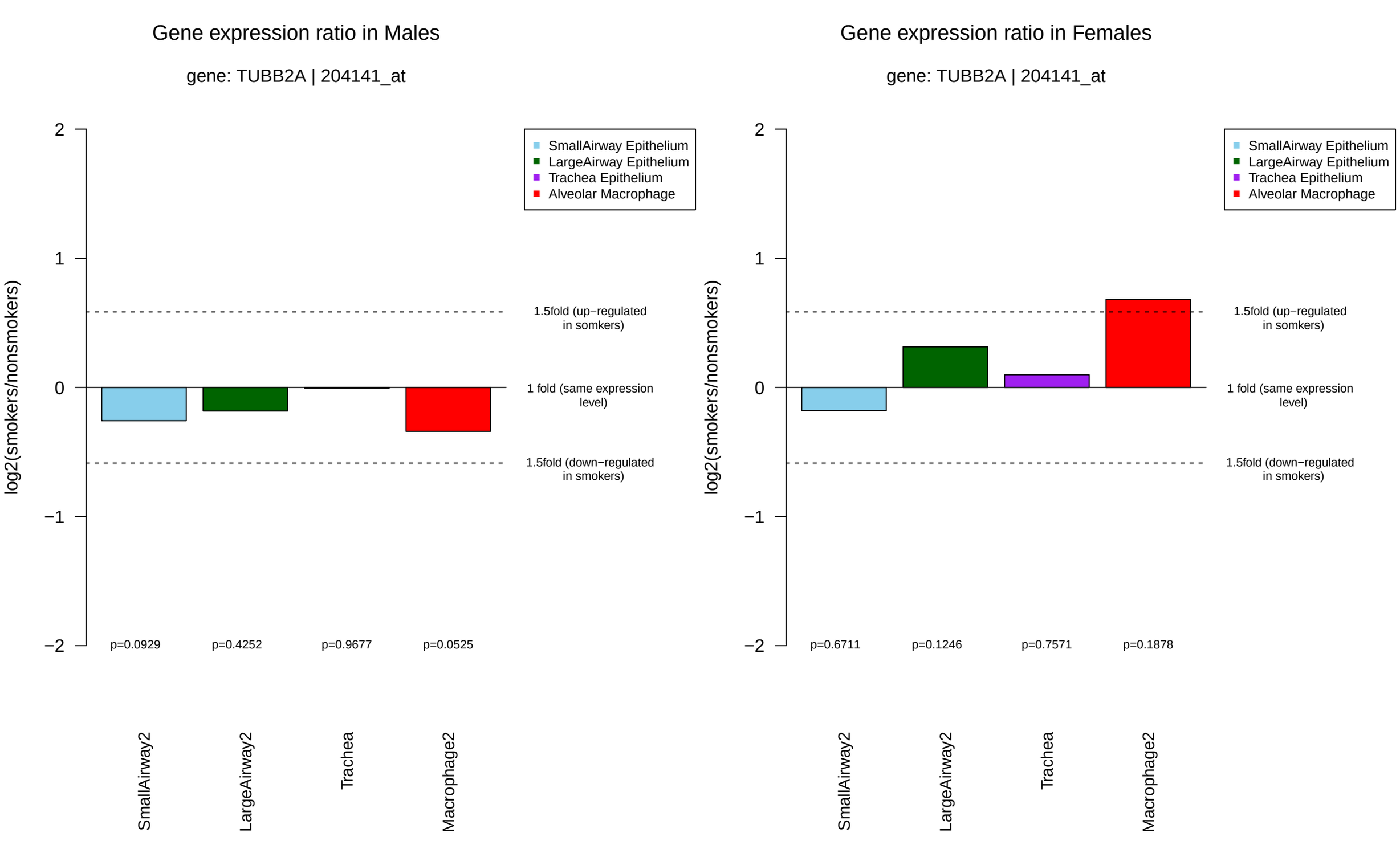
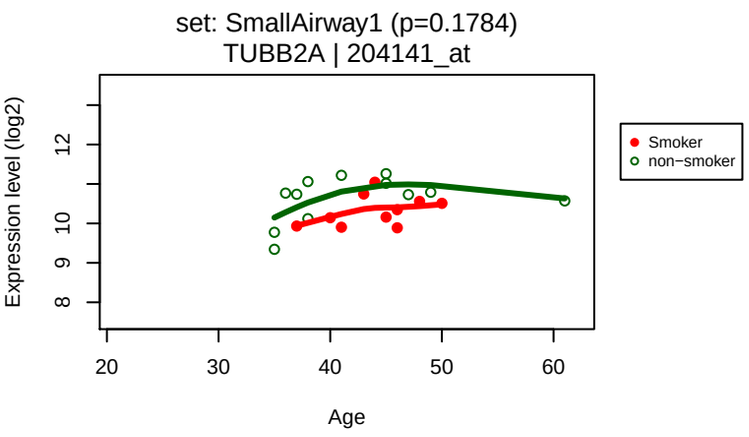
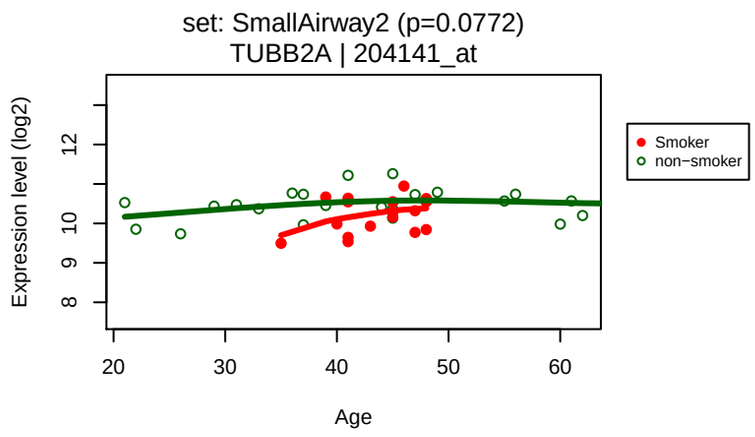
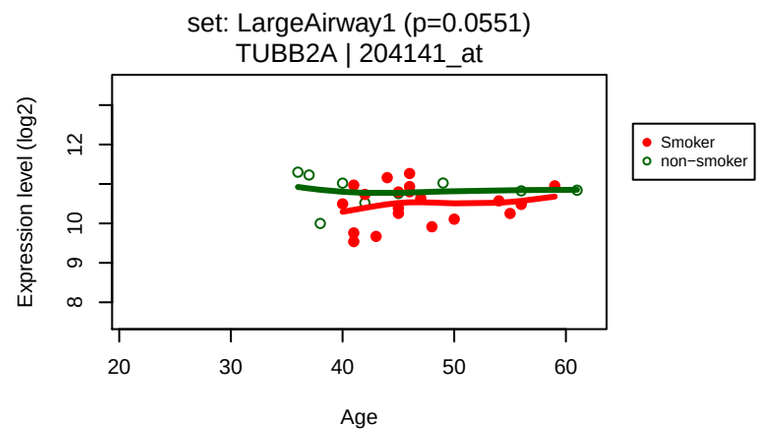

Gene expression of smokers and nonsmokers
gene: TUBB2A | 204141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1784
p=0.0772
p=0.63
p=0.0551
p=0.9586
p=0.7898
p=0.1092
p=0.9321
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

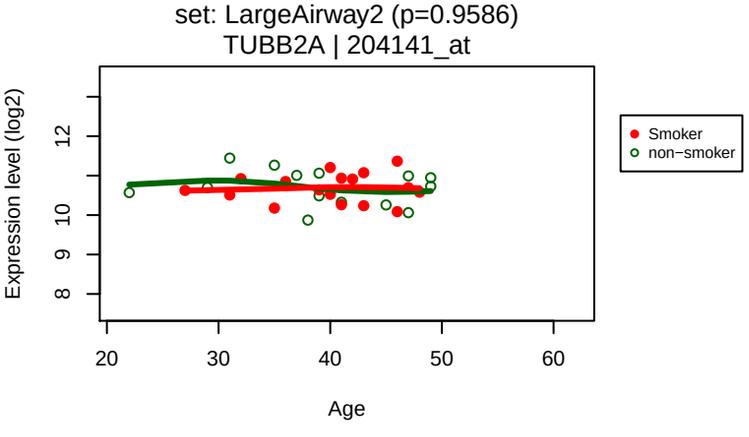

Gene expression ratio (smokers/nonsmokers)
gene: TUBB2A | 204141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1784
p=0.0772
p=0.63
p=0.0551
p=0.9586
p=0.7898
p=0.1092
p=0.9321
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

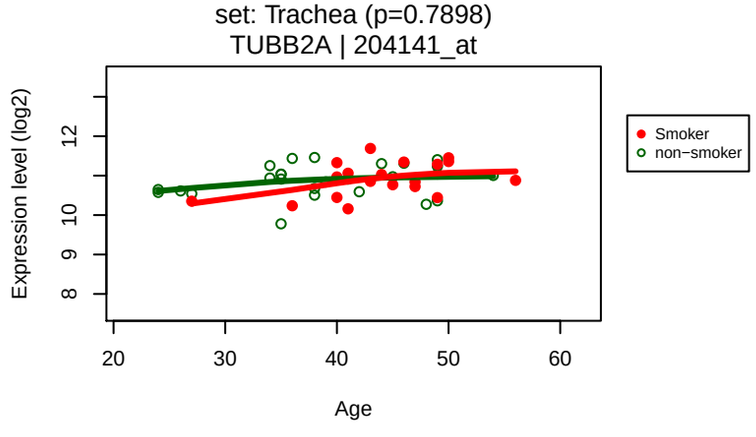

Gene expression ratio in Males
gene: TUBB2A | 204141_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0929
p=0.4252
p=0.9677
p=0.0525
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TUBB2A | 204141_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6711
p=0.1246
p=0.7571
p=0.1878
SmallAirway2
LargeAirway2
Trachea
Macrophage2

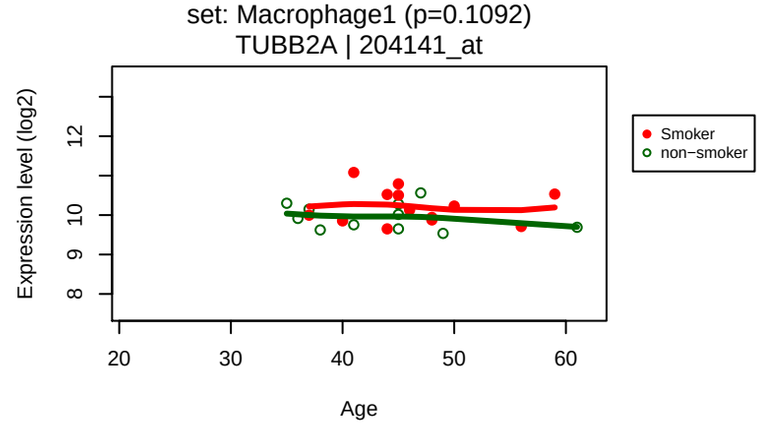

set: SmallAirway1 (p=0.1784)
TUBB2A | 204141_at
Expression level (log2)
Age
Smoker
non−smoker

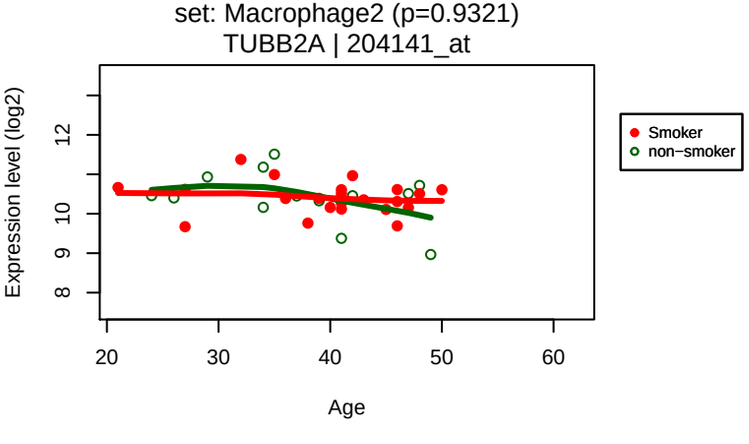

set: SmallAirway2 (p=0.0772)
TUBB2A | 204141_at
Expression level (log2)
Age
Smoker
non−smoker

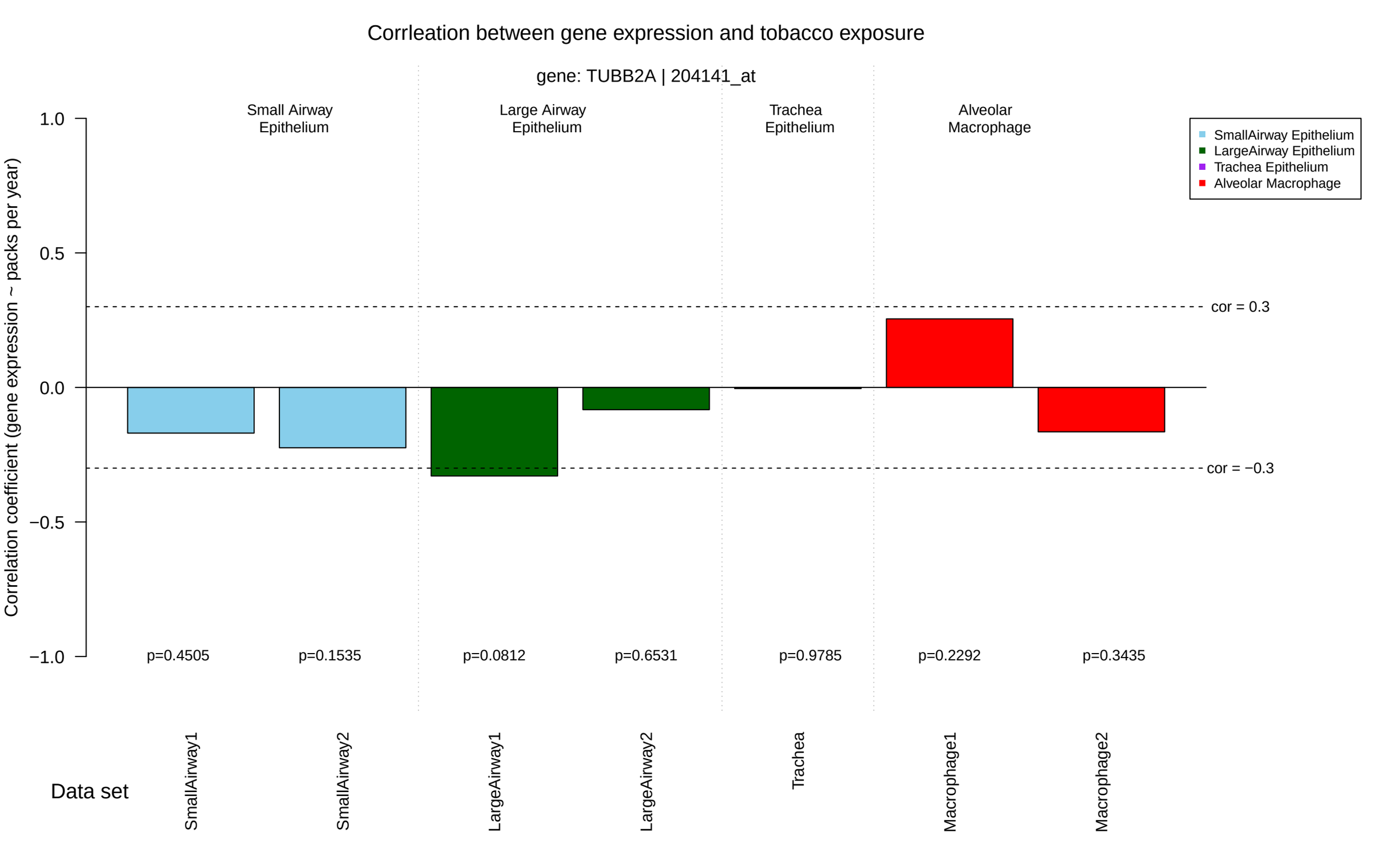

set: LargeAirway1 (p=0.0551)
TUBB2A | 204141_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9586)
TUBB2A | 204141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7898)
TUBB2A | 204141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1092)
TUBB2A | 204141_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9321)
TUBB2A | 204141_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TUBB2A | 204141_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4505
p=0.1535
p=0.0812
p=0.6531
p=0.9785
p=0.2292
p=0.3435
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2